BASEL AHMAD SHABAB

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EDUCATION

Ph.D. in Mining and Minerals Engineering **Spring 2024 – Present**

Virginia Tech, Blacksburg, USA (Expected Graduation: Fall 2027)

Research Area: Experimental investigation of cementitious materials durability and reactive transport behavior in underground hydrogen storage systems.

Relevant Courses: Characterization Techniques for Cement-Based Materials, Multicomponent Thermodynamics, Microfluidics and Nanofluidics, Numerical Modeling of Petroleum Reservoirs.

B.S. in Petroleum and Mining Engineering **Spring 2016 – Summer 2021**

Chittagong University of Engineering & Technology (CUET), Chittagong, Bangladesh

TECHNICAL SKILLS

- **Laboratory Tools:** SEM-EDS, XRD, XRF, TGA, micro-CT imaging, Tensile Strength, Helium Pycnometry, Gas Permeametry
- **Programming and Data Analysis:** Python (Data analytics), Rocscience RS2, Profex, OriginPro, ImageJ, MS Office
- **Safety Trainings:** Compressed Gas Cylinder, Hazard Assessment, Lock-out Tag-out, Respiratory Protection

WORK EXPERIENCE

Graduate Research Assistant **Spring 2024 – Present**

Department of Mining & Minerals Engineering, Virginia Tech

- Investigated durability and sealing performance of Class H wellbore cement exposed to hydrogen gas and reservoir brines under high-pressure, elevated temperature conditions relevant to underground hydrogen storage.
- Conducted multi-scale geomechanical, geochemical, and petrophysical characterization of Class H cement using SEM-EDS, XRD, XRF, strength and porosity/permeability analyses to link microstructural evolution to macroscale transport and mechanical behavior.
- Quantified fracture behavior and elastic properties (Young's modulus, Poisson's ratio) of cemented paste backfill (CPB) via S- and P-wave velocity measurements to support mine design optimization.

Engineering Economics Intern **Fall 2024**

Society of Exploration Geophysicists (SEG)

- Evaluated geological and geochemical suitability of offshore Texas formations for long-term CO_2 storage, integrating petrophysical data with reservoir-scale economic modeling and risk analysis.
- Designed a measurement, monitoring, and verification (MMV) plan as part of the multidisciplinary CCUS feasibility assessment; collaborated with a 6-member team spanning geoscience, engineering, and economics.

KEY PROJECTS

Wellbore Integrity in Underground Hydrogen Storage - Depleted Gas Fields of Appalachia **Fall 2024**

- Established baseline experimental data on Class H cement degradation under hydrogen-brine exposure, addressing a critical knowledge gap in subsurface hydrogen storage infrastructure design.

Backfill Fracturing Tool for In-Situ Strength Estimation **Spring 2025**

- Characterized the mechanical properties of cemented paste backfill materials.

PUBLICATIONS

- **Shabab, B. A., &** Pandey, R. (2026). ***Experimental evaluation of wellbore cement integrity in underground hydrogen storage environments***. International Journal of Hydrogen Energy. [DOI]: https://doi.org/10.1016/j.ijhydene.2026.153957
- Frimpong, J. A., **Shabab, B. A.**, et al. (2025). ***Fracture initiation pressure as a measure of cemented paste backfill strength.*** Mining, Metallurgy & Exploration. [DOI]: https://doi.org/10.1007/s42461-025-01257-6

HONORS & AWARDS

- Pratt Fellowship, Virginia Tech (Fall 2024 – Spring 2025)
- Academic Merit Scholarship - CUET (Top 10% ranking for 7 out of 8 semesters)

LEADERSHIP EXPERIENCE

- Logistics Secretary, Association for Bangladeshi Students at Virginia Tech **Summer 2025 – Present**
- Student Member, VT SPE, SME, and SEG Student Chapters **Spring 2024 – Present**
- Vice President , CUET SPE Student Chapter **Spring 2019 – Summer 2020**